

STATISTICS FOR DATA ANALYTICS
APPLICATION ASSIGNMENT
50-Row E-Commerce / Retail Dataset

DATA UNDERSTANDING :

➤ Identify all numerical variables:

- ✚ Age
- ✚ Order_Value
- ✚ Items_Purchased
- ✚ Discount_Percent
- ✚ Delivery_Days
- ✚ Customer_Rating
- ✚ Marketing_Spend

➤ Identify all categorical variables.

- ✚ Gender
- ✚ Repeat_Customer
- ✚ Payment_Mode
- ✚ Order_Value_Band
- ✚ Customer_ID

➤ Identify which numerical variables are discret

- ✚ Items_Purchased
- ✚ Delivery_Days
- ✚ Customer_Rating

➤ Identify which numerical variables are Continuous

- ✚ Age
- ✚ Order_Value
- ✚ Discount_Percent
- ✚ Marketing_Spend

➤ Check for missing values.

- ✚ Missing Values Not Presented

➤ Check for duplicate Customer_ID values.

- ✚ Duplicate Customer Id Not Appeared

➤ Identify the variable that represents

customer spending	Order_Value
Customer satisfaction	Customer_Rating
Delivery performance	Delivery_Days
Marketing investment	Marketing_Spend

DATA DICTIONARY :

Column	Data Type	Variable Type	Business Meaning
Customer_ID	Text	Identifier	Unique customer identification
Age	Integer	Continuous	Customer age
Gender	Text	Categorical	Customer gender
Order_Value	Integer	Continuous	Customer spending
Items_Purchased	Integer	Discrete	Number of items purchased
Discount_Percent	Integer	Continuous	Discount offered
Delivery_Days	Integer	Discrete	Delivery duration
Customer_Rating	Integer	Discrete	Customer satisfaction
Marketing_Spend	Integer	Continuous	Marketing investment
Repeat_Customer	Text	Categorical	Marketing investment
Payment_Mode	Text	Categorical	Payment method
Order_Value_Band	Text	Categorical	Spending category

2. MEAN, MEDIAN & MODE :

- Use order value calculate the mean
📊 MEAN IS **6939 Rs**
- Use order value calculate the median
📊 MEDIAN IS **5500Rs**
- Use order value to Find the mode, if applicable.
📊 MODE IS Not Applicable
- Compare mean and median.
📊 Mean Is Greaterthen Median
- Typical customer order value
📊 Median – 5500Rs
- Explain Why.
📊 Median Is Not Affect Largely.

➤ Challenge

✚ typical customer spend is Median -5500Rs Because Middle Value Affected Very Low.Outlier Issue Low.

3. VARIANCE & STANDARD DEVIATION :

➤ Use Order_Value calculate the variance.

✚ VARIANCE IS **29088229**

➤ Calculate standard deviation.

✚ STANDARD DEVIATION IS **5393.35 Rs**

➤ Is customer spending tightly concentrated or widely spread?

✚ Customer spending is Widely Spread.

➤ What does the standard deviation tell you?

✚ Customer Spending shows more variance compare to mean vale.

➤ Why is standard deviation easier to communicate than variance?

✚ Because Two Are Same Units.

➤ Does a high standard deviation necessarily mean the data is bad?

✚ No

➤ Business Question:

✚ If have higher standard deviation its have higher variance.

4. QUARTILES, PERCENTILES & IQR:

➤ Calculate Q1

✚ Q1 IS **3587.50 Rs**

➤ Calculate Q2

✚ Q2 IS **5500 Rs**

➤ Calculate Q3

✚ Q3 IS **8325 Rs**

➤ Calculate IQR

✚ IQR IS **4737.50 Rs**

➤ Calculate P25, P50, P75 and P90

✚ P25 IS **3587.50 Rs**

✚ P50 IS **5500 Rs**

✚ P75 IS **8325 Rs**

✚ P90 IS **12650 Rs**

- What does Q1 mean?
 - ✚ 25% of customers have Order_Value below 3587.50 Rs.
- What does Q3 mean?
 - ✚ 75% of customers have Order_Value below 8325 Rs.
- What percentage of observations lie below Q3?
 - ✚ 75% of observations lie below Q3.
- What does P90 mean?
 - ✚ 90% of customers have Order_Value below 12650 Rs.
- What does IQR tell us about customer spending?
 - ✚ spread of the middle 50% of customer spending.
- Business Challenge
 - ✚ I would use P90.

5. OUTLIER INVESTIGATION

- Calculate Q1, Q3 and IQR
 - ✚ Q1 IS 3587.50 Rs
 - ✚ Q3 IS 8325 Rs
 - ✚ IQR IS 4737.50 Rs
- Calculate Lower Boundary and Upper Boundary
 - ✚ LOWER BOUNDARY IS -3518.75 Rs
 - ✚ UPPER BOUNDARY IS 15431.25 Rs
- Identify potential outliers
 - ✚ 15500 Rs
 - ✚ 18000 Rs
 - ✚ 22000 Rs
 - ✚ 30000 Rs
- Are these observations necessarily errors?
 - ✚ No
- Give possible business reasons for a very high order.
 - ✚ Bulk Purchase
 - ✚ Premium Products
- Calculate the mean before investigating the outlier.
 - ✚ MEAN BEFORE OUTLIER IS 6939 Rs
- Recalculate the mean after excluding potential outliers.
 - ✚ MEAN AFTER OUTLIER IS 5683.70 Rs
- How much did the mean change?
 - ✚ Mean decreased by 1255.30 Rs.
- Should a Data Analyst automatically delete an outlier?
 - ✚ No

6. HISTOGRAM & BOXPLOT:

- What does the shape look like?
 - ✚ right-skewed
- Where are most orders concentrated?
 - ✚ Most orders are concentrated between ₹850 and ₹11,250.
- Is the distribution symmetric?
 - ✚ No
- Does it appear right-skewed or left-skewed?
 - ✚ It appears right-skewed
- Are there extreme values?
 - ✚ Yes, Some customers spend much more than most customers, especially above ₹15,000
- Identify the median.
 - ✚ Median is 5500 Rs.
- Identify Q1 and Q3.
 - ✚ Q1 is 3587.50 Rs and Q3 is 8325 Rs.
- Identify the IQR.
 - ✚ IQR is 4737.50 Rs.
- Identify potential outliers.
 - ✚ Potential outliers are 15500 Rs, 18000 Rs, 22000 Rs and 30000 Rs.
- Business Question
 - ✚ I would use a Boxplot because it clearly shows potential outliers separately from the main data.

7. SKEWNESS & KURTOSIS AWARENESS

- Order_Value distribution is symmetric, right-skewed or left-skewed.
 - ✚ The Order_Value distribution is Right-Skewed.
- Explain what the skewness tells you about customer spending.
 - ✚ that most customers have lower or medium spending, few customers have very high spending.
- Check the kurtosis using Python.
 - ✚ I Using excel To KURTOSIS IS 6.393
- State whether the distribution shows heavier or lighter tails compared with a normal distribution.
 - ✚ The distribution shows relatively heavier tails compared with a normal distribution.
- Business Interpretation
 - ✚ Most customers make small or medium-value purchases, while a few customers make very high-value purchases. This creates a right-skewed distribution.

8. PROBABILITY

- Probability that a randomly selected customer is a Repeat Customer
 - ✚ PROBABILITY IS 0.32 OR 32%
 - ✚ There is a 32% probability that a randomly selected customer is a Repeat Customer.
- Probability that an order has a discount greater than 20%.
 - ✚ PROBABILITY IS 0.28 OR 28%
 - ✚ There is a 28% probability that a randomly selected order has a discount greater than 20%.
- Probability that an order takes more than 5 days to deliver.
 - ✚ PROBABILITY IS 0.10 OR 10%
 - ✚ There is a 10% probability that a randomly selected order takes more than 5 days to deliver.

9. CONDITIONAL PROBABILITY

- Repeat Customer given Order_Value > ₹5,000
 - ✚ PROBABILITY IS 46.43%
- Calculate the overall probability of being a Repeat Customer.
 - ✚ OVERALL PROBABILITY IS 32%
- Compare the two probabilities.
 - ✚ Conditional Probability is 46.43% and Overall Probability is 32%.
- Which probability is higher?
 - ✚ Conditional Probability is higher.
- What does the difference suggest?
 - ✚ Order_Value greater than ₹5,000 have a higher chance of being Repeat Customers.
- Does this prove that high spending causes repeat purchases?
 - ✚ No
- Give one possible factor other than Order_Value that could influence repeat purchasing.
 - ✚ Customer satisfaction can influence repeat purchasing.

10. RANDOM VARIABLES

- Identify two discrete random variables from the dataset.
 - ✚ Items_Purchased
 - ✚ Delivery_Days
- Identify two continuous random variables from the dataset.
 - ✚ Order_Value
 - ✚ Marketing_Spend
- Explain why you classified each variable that way.
 - ✚ Items_Purchased and Delivery_Days are discrete because they take countable whole number values.
 - ✚ Order_Value and Marketing_Spend are continuous because their values can be measured and can take different values.
- Define $X = \text{Order_Value}$ of the selected customer. Explain what possible values X can take.
 - ✚ X represents the Order_Value of a randomly selected customer.
 - ✚ X can take any Order_Value from the dataset,
 - ✚ such as 850 Rs, 1200 Rs, 2350 Rs, 5500 Rs, 18000 Rs or 30000 Rs.
- Define $Y = \text{Items_Purchased}$. Explain how Y differs from X as a random variable.
 - ✚ Y represents the number of items purchased by a customer.
 - ✚ Y takes countable whole number values, while X represents the customer Order_Value.

11. SAMPLING & SAMPLING BIAS

- How would you select the sample?
 - ✚ I would randomly select 500 customers from the 50,000 customers.
- Would you use probability or non-probability sampling?
 - ✚ I would use Probability Sampling.
- Which sampling technique would you recommend?
 - ✚ I recommend Simple Random Sampling.
- Why?
 - ✚ It gives every customer an equal chance of being selected and helps reduce sampling bias.
- What could go wrong if the sample is not representative?
 - ✚ The survey results may not correctly represent the overall customer population.

- Is this necessarily representative of all customers?
 - ✚ No
- Who might be missing?
 - ✚ Customers who did not purchase during the last 7 days may be missing.
- What type of bias could occur?
 - ✚ Selection Bias could occur.
- How would you improve the sampling approach?
 - ✚ I would select customers from different time periods instead of selecting only recent customers.

- Identify the sampling approach and discuss its limitation.
 - ✚ This is Convenience Sampling.
 - ✚ It selects customers who are easiest to contact, but it may not represent the overall customer population and can cause sampling bias.